Relaytic-AML: A Local-First Agentic Evaluation Lab for Financial-Crime Machine Learning

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Abstract

Anti-money laundering (AML) machine-learning experiments are hard to audit when data residency, temporal validity, graph provenance, leakage control, agent assistance, review capacity, and public claims live in separate tools. Relaytic-AML is a local-first agentic evaluation lab: role-scoped agents and deterministic harnesses turn local runs into evidence cells, then claim gates decide which public interpretations those cells may support. The current evidence pack exercises this substrate on PaySim, Elliptic, and Elliptic2 stress cases, with PR-AUC rows of 0.6388, 0.6688, and 0.9432 +/- 0.0009, respectively; the Elliptic2 row is benchmark context against a RevClassifyDS reference of 0.9740. The contribution is a reproducible governance architecture for local AML experimentation, rowless external-agent handoff, artifact-backed release checks, and public wording that stays tied to recorded evidence roles.

1 Introduction

AML modeling is a rare-event, high-stakes evaluation problem. The input may be a stream of mobile-money transfers, card events, account activity, wire messages, customer profiles, or blockchain transactions. Suspicion rarely appears in one row. Suspicion more often appears as a temporal or network pattern: fast movement of funds after receipt, repeated transfers through related accounts, structured amounts, new counterparties receiving many payments, activity that does not match a customer's profile, or cryptocurrency behavior that involves risky services and geography. Regulatory and typology material from FATF, FinCEN, and FFIEC describes this kind of pattern-based reasoning in operational language [Financial Action Task Force, 2020, Financial Crimes Enforcement Network, 2026, Federal Financial Institutions Examination Council, 2026].

Because AML suspicion is often pattern-based, isolated AML model metrics are easy to misread. A PR-AUC value or a precision-at-review-budget number becomes interpretable only when the evaluation record shows which data stayed local, which fields were available before the decision, how temporal or graph boundaries were split, whether model selection touched the test surface, which review capacity was assumed, and which claim boundary the metric can support. Agent assistance raises the standard for provenance rather than lowering it: fluent explanations from large language models (LLMs) or coding agents can drift away from the artifact record unless the system is designed to fail closed.

Relaytic began as a general local-first inference-engineering lab. Relaytic-AML is the financial-crime edition used here to test whether that architecture can support governed AML experimentation. It is a set of cooperating agents and deterministic harnesses around a local artifact store. The guide helps a user or another agent understand where the run is. The scout checks source posture, schema, leakage risk, and split feasibility. The strategist turns the objective into a task contract. The scientist challenges baselines, ablations, and budget choices. The builder executes bounded runs. Reviewers reconstruct traces. Release governors lint claims, figures, tables, source packages, and public wording against the evidence record.

Relaytic-AML contributes a local-first evidence and release-governance layer for AML machine-learning experiments, not a new detector architecture. The benchmark rows matter because they exercise the architecture under temporal, graph, operating-point, and claim-governance pressure. The central question is whether a local evaluation lab can keep evidence, agent assistance, privacy boundaries, and publishable claims aligned while preserving the role of each result.

The work is organized around four research questions:

- **RQ1:** Can Relaytic-AML produce reproducible evidence cells for AML-style temporal and graph tasks?
- **RQ2:** Can it prevent leakage-prone or unsupported claims from being promoted?
- **RQ3:** Can the local artifact record support rowless handoff to external agents while preserving provenance?
- **RQ4:** Do the benchmark rows demonstrate useful evaluation evidence under explicit split and budget contracts?

The paper makes three contributions. First, it presents a local-first evidence-cell model for AML experiments that binds each reported number to dataset, split, command, artifact, budget, leakage posture, and operating point. Second, it implements deterministic release gates that block leakage-prone interpretations, unsupported public wording, and unsafe external-agent handoff. Third, it evaluates the architecture through PaySim, Elliptic, and Elliptic2 stress cases plus deterministic fixtures for temporal validity, graph provenance, benchmark-context handling, handoff, recovery, and claim governance.

2 Related Work

The AML benchmark landscape is moving toward larger, more realistic, and more graph-native settings. PaySim remains useful as a synthetic mobile-money fraud simulator for temporal proxy experiments [Lopez-Rojas et al., 2016]. Elliptic introduced a public Bitcoin transaction graph with anonymized node features and temporal labels [Weber et al., 2019]. Elliptic2 and RevTrack/RevClassify shifted attention to suspicious subgraphs and richer blockchain context [Bellei et al., 2024, Song et al., 2024]. Recent work such as TransXion, LineMVGNN, quasi-temporal graph extraction, BlazingAML, and continual graph-learning studies shows that the research frontier increasingly treats AML as dynamic graph and systems work rather than static tabular classification [Chen et al., 2026, Poon et al., 2026, Tariq and Hassani, 2026, Ye et al., 2026, Deprez et al., 2025].

Detector papers such as TransXion, BlazingAML, LineMVGNN, and RevClassifyDS push the model frontier. Relaytic-AML sits one layer around that work: it asks how experiments should be governed when data is local or licensed, the task may be temporal or graph-based, analyst review capacity matters, and an agent-assisted workflow must still produce evidence that a skeptical reviewer can audit. The focus on governed local experimentation places Relaytic-AML near dataset documentation, model reporting, reproducibility practice, experiment tracking, and governance work [Gebru et al., 2021, Mitchell et al., 2019, Pineau et al., 2021, Zaharia et al., 2018].

Experiment-tracking systems preserve runs and artifacts [Zaharia et al., 2018]; model cards describe trained models; datasheets describe data; reproducibility checklists improve reporting; and agent benchmarks evaluate agent behavior. Relaytic-AML handles a different responsibility: determining which public scientific claims a local AML evidence cell is allowed to support. The claim-governance responsibility is especially important when evidence comes from licensed files, proxy datasets, temporal graph tasks, or rowless handoff packets rather than from a single open leaderboard run.

A closely related newer line uses LLMs and agents for AML triage, graph-context reasoning, suspicious activity report (SAR) narrative support, compliance serving stacks, and runtime agent governance [Pirmorad, 2025, Naik et al., 2025, 2026, Gaurav et al., 2025, Kaptein et al., 2026]. Those systems make agent assistance more capable, but they also make evidence boundaries more important. Relaytic-AML is not a SAR drafting system, not an LLM detector, and not a general-purpose agent-governance product. Its role is narrower: keep local AML evidence, rowless handoff, and public claims aligned.

What is new. Relaytic-AML is a governance substrate around detectors and agent-assisted workflows. It would be used to wrap detector outputs, LLM explanations, hosted score files, and paper tables with evidence cells, rowless handoff, and claim gates. A company would not use it as a detector replacement; it would use it to make the local evaluation record inspectable before a result becomes a benchmark row, a handoff packet, or a public claim. In short, Relaytic-AML is not a detector replacement. It is the local evidence layer around detectors and agents.

Table 1: Adjacent systems comparison.

Family	Primary object	Relaytic-AML position	Boundary
Model cards and model reporting [Mitchell et al., 2019]	trained model and intended-use report	adds command-level metric provenance, release gating, and stronger-claim blocking around AML experiments	does not replace model reporting
Datasheets and dataset documentation [Gebru et al., 2021]	dataset creation, composition, collection, and recommended use	connects dataset posture to split contracts, leakage controls, benchmark rows, and admissible claims	does not replace source-dataset governance
ML reproducibility checklists [Pineau et al., 2021]	reporting checklist and reproducibility discipline	turns checklist-like obligations into executable artifact-generation and release gates	does not claim full independent reproduction for licensed data
MLOps experiment tracking [Zaharia et al., 2018]	runs, metrics, parameters, artifacts, lineage, and model versions	focuses on local AML evidence, privacy posture, rowless handoff, and public scientific claim admissibility	is not a hosted tracker or production model registry
Agent benchmarks and research-agent evaluations [Chen et al., 2025, Starace et al., 2025]	agent performance on research, coding, or skill-use tasks	uses agents inside a governed local evaluation lab and then tests whether their outputs stay artifact-attached	does not benchmark a general-purpose agent
AML detector and benchmark papers [Weber et al., 2019, Bellei et al., 2024]	detector architecture, benchmark result, graph construction, or financial-crime dataset	provides the local evidence and claim-governance substrate that such detector studies can run through	governs detector evidence rather than introducing a graph-neural model
AML LLM graph reasoning and triage systems [Pirmorad, 2025, Naik et al., 2026]	LLM reasoning, triage, serving, and evidence-rich prompts for AML workflows	keeps LLM or external-agent help downstream of rowless local evidence, artifact provenance, and claim gates	does not claim an LLM detector or AML LLM-serving stack
Agentic SAR and compliance narrative assistants [Naik et al., 2025]	human-in-the-loop SAR or compliance narrative drafting	governs the local experimental evidence and admissible claims that such narrative workflows should cite	does not generate or validate regulatory SAR submissions
Agent governance and runtime trust layers [Gaurav et al., 2025, Kaptein et al., 2026]	runtime policies, enforcement, logging, trust scoring, and path-dependent agent governance	specializes governance to local AML evidence cells, rowless handoff, benchmark context, and paper/public claim a...	does not claim to be a general-purpose agent-governance product

Relaytic-AML does not replace dataset documentation, model cards, experiment trackers, or detector papers. The system ties those concerns together for local AML research: a model result is only reader-facing after its source posture, split, leakage policy, budget, artifact field, handoff posture, and claim boundary are visible.

Recent agent-evaluation work shows that language-model systems can produce persuasive research artifacts while still failing on reproduction, validation, or expert-level judgment [Chen et al., 2025, Starace et al., 2025, Wijk et al., 2025]. Skill- and tool-using agents make that opportunity larger and the governance problem sharper [Yang et al., 2026]. Relaytic-AML responds by making model scores, public claims, handoff packets, and paper assets downstream of local artifacts rather than downstream of a conversation transcript.

3 System Overview

Relaytic-AML is built around one authority rule: truth-bearing records live in the local workspace, not in the conversation. Raw data, licensed benchmark files, run summaries, traces, metric cells, model outputs, tables, figures, and release reports live on disk. Agents may explain, propose, and repair, but their proposals only become evidence when they are materialized as artifacts another human or agent can inspect.

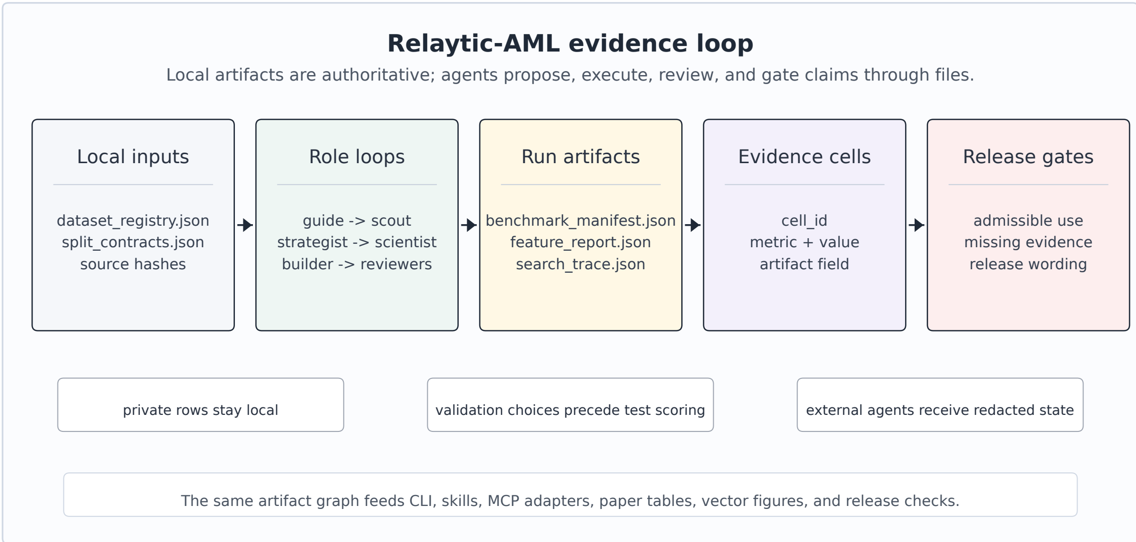

Figure 1: Relaytic-AML local-first architecture: local data and artifacts flow through role-scoped agents into evidence cells, interpretation gates, and paper/release/handoff surfaces.

Figure 1 summarizes the local evidence loop. Dataset registries and split contracts enter the role-scoped agent runtime. Candidate runs write benchmark manifests, search traces, feature reports, and metric cells. Claim gates read those cells together with release audits and emit only the interpretations that the evidence supports. The same contract feeds the command-line interface, project skills, OpenClaw-style handoff, Claude/Codex skill files, and Model Context Protocol (MCP) adapters.

The external-agent path is rowless by default. Relaytic can export current state, next-action options, artifact shortlists, and safe commands for another model without sending raw transaction rows, secrets, or private local paths. A local LLM can optionally help phrase guidance, but the deterministic guide and evidence artifacts remain the source of truth. The separation between advisory help and local evidence matters for private AML work: outside intelligence may help navigate the run, but data residency and claim provenance stay local unless an operator deliberately changes policy.

4 Evidence Cell and Claim-Gate Design

An evidence cell is the unit that makes a paper number auditable. Rather than storing a bare metric value, it records the dataset, split, command, artifact field, model or feature budget, leakage posture, and operating point. Interpretation is deliberately stored in a separate gate output: the cell says what happened, and the gate says how that fact may be used.

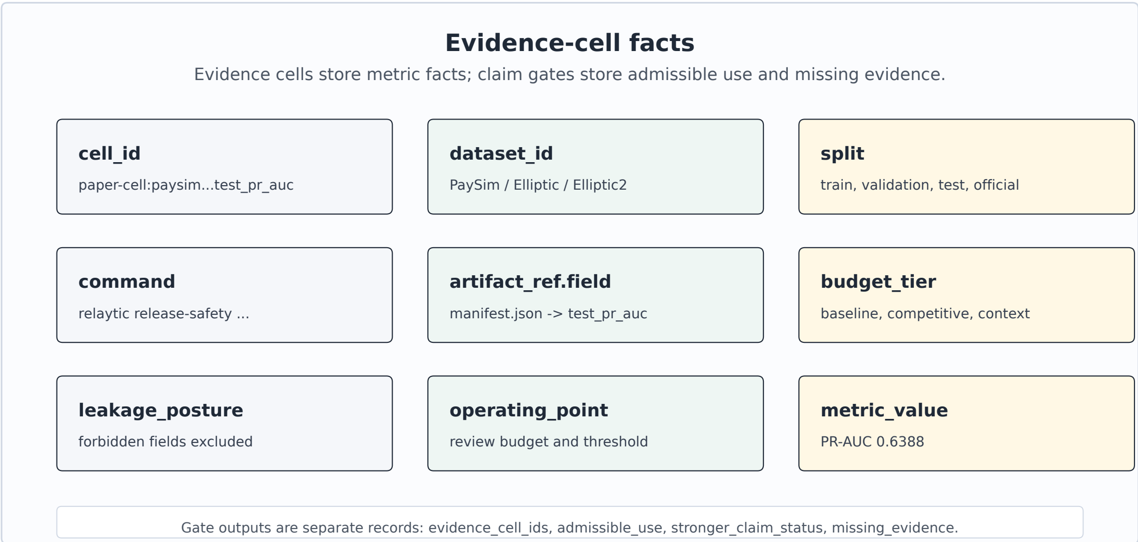

Figure 2: Evidence-cell schema: every reported number carries dataset, split, command, artifact, budget, leakage posture, operating point, metric, and value; interpretation is stored separately.

The table below uses compact publication aliases for readability; the full machine metric-cell identifiers are preserved in the metric-cell audit artifact and generated table comments. Keeping the factual metric record separate from the claim boundary is the central design choice.

Table 2: Representative evidence cells.

ID	Dataset	Metric	Value	Split	Artifact	Evidence role
PS-PR	PaySim	test PR-AUC	0.6388	temporal test	PaySim run; manifest	bounded demonstration
PS-P@B	PaySim	precision at review budget	0.7033	temporal test	PaySim run; manifest	bounded demonstration
EL-PR	Elliptic	test PR-AUC	0.6688	graph-time test	graph run; feature table	graph-feature evidence
EL-P@B	Elliptic	precision at review budget	1.0000	graph-time test	graph run; feature table	graph-feature evidence
E2-PRm	Elliptic2	official test PR-AUC mean	0.9432	official test	E2 run; scorecard	external reference/context
E2-ref	Elliptic2	published reference PR-AUC	0.9740	reported ref.	E2 run; scorecard	external reference/context

A representative record is compact enough to audit directly. The public table uses the alias `PS-PR`, while the underlying artifact keeps the longer machine identifier. The example shows the factual record that the claim gate later consumes; stronger interpretations are deliberately kept outside the cell.


```
{
  "cell_id": "PS-PR",
  "dataset_id": "paysim_temporal_transaction_fraud",
  "split": "temporal test",
  "command": "paysim-competitive --budget-tier competitive",
  "artifact_ref": "paper_metric_cell_audit:test_pr_auc",
  "metric": "test_pr_auc", "value": 0.6388,
  "leakage_posture": "balance and raw IDs excluded",
  "claim_state": "bounded PaySim proxy; stronger claims need holdout"
}
```


Algorithm 1 Evidence-cell creation

Input: dataset registry D, split contract S, candidate budget B, run artifacts A
Output: evidence cell c with factual provenance
Freeze source posture, license posture, task target, and split contract.
Derive only features allowed by S and record excluded leakage fields.
Run baseline candidates under the declared baseline budget.
Run stronger candidates only within B and select on validation evidence.
Evaluate the selected row once on the fixed test surface.
Write c = dataset, split, command, artifact field, metric, value, budget, leakage posture, and operating point.
Hand c to the claim gate before it appears in tables, figures, or release text.

The claim gate is the second half of the design. Its job is conservative by construction: if the evidence cell is incomplete, if a split is leakage-prone, if a metric is only a proxy, or if a stronger interpretation needs a different dataset or study, the gate preserves the evidence and routes the stronger use to an evidence-needs record. The gate is implemented as a release mechanism, so it changes what the paper artifact pipeline and public release surfaces are allowed to say.

Algorithm 2 Claim-gate validation

1: **Input:** public claim q, evidence cells C, gates G, limitations L
2: **Output:** admissible wording and evidence-needs record
3: Resolve every evidence cell named by q and require dataset, split, command, artifact, budget, and leakage fields.
4: Compare the strength of q with source posture, split validity, metric scope, and benchmark role.
5: If q is exactly supported, emit the admissible wording and the evidence-cell identifiers.
6: If q is stronger than C and G permit, record the stronger-claim status and gate reason.
7: Attach the missing evidence needed to make q testable in future work.
8: Route current evidence to its admissible paper use and keep stronger uses out of headline wording.

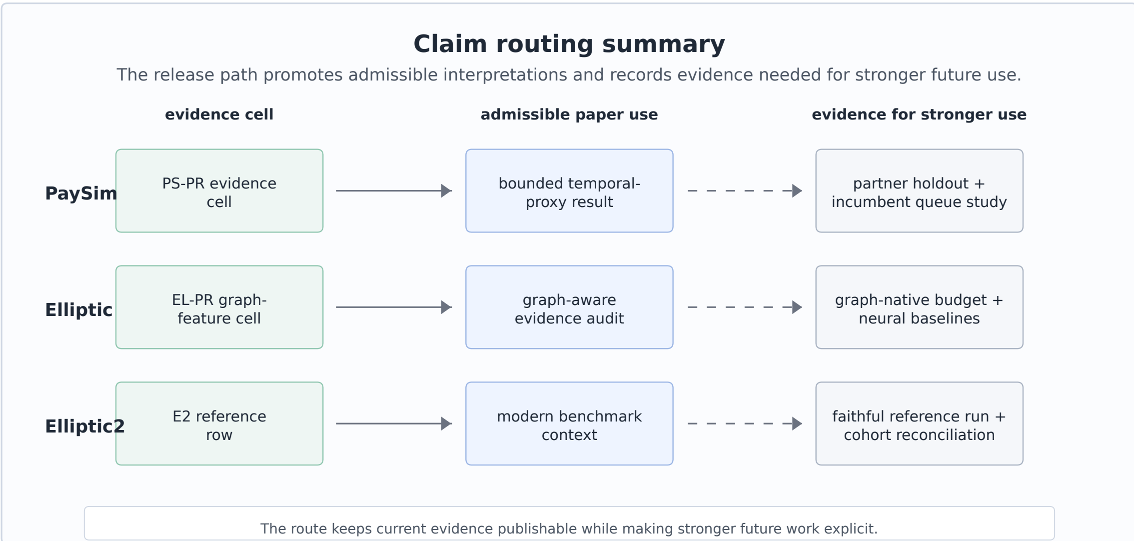

Figure 3: Claim routing summary: current cells map to admissible paper uses and to evidence needed for stronger future interpretations.

Figure 3 gives concrete routing behavior. A PaySim row becomes a temporal-proxy demonstration, an Elliptic row becomes graph-feature evidence with temporal provenance, and an Elliptic2 row becomes external benchmark context. The same records also specify what evidence would be needed before stronger future uses could be made.

5 Experimental Protocol

The experiments test Relaytic-AML as an evaluation lab. The datasets are separated by evidence role. PaySim is the main empirical demonstration because it is fully local and supports a controlled temporal proxy workflow. Elliptic tests temporal graph provenance and feature-view discipline. Elliptic2 tests whether the system can keep a modern external reference row visible without overstating its role.

The table below records the context a reader needs before interpreting any metric. The positive rates show why precision-recall area under the curve (PR-AUC) is the primary score. These are rare-event tasks where receiver-operating-characteristic area under the curve can look strong while the review queue remains poor. The split rules are equally important: PaySim is split by chronological simulator step, Elliptic by time-step graph windows, and Elliptic2 by the official/context partitions recorded in the local evidence pack.

Table 3: Dataset, split, and feature policy.

Dataset	Scale and split	Feature policy	Metrics	Evidence role
PaySim synthetic mobile-money transaction fraud	Synthetic temporal transaction fraud; 6,362,620 rows/nodes; 8213 positives; time step	26 decision-time features; balance columns excluded	PR-AUC, P@k, R@budget	bounded demonstration
Elliptic Bitcoin transaction graph	Temporal Bitcoin graph node classification; 203,769 rows/nodes; 4545 positives; graph time	structural same snapshot only; source provided flattened features; source features plus structural snapshot	PR-AUC, P@k, fixed-FPR recall	graph-feature evidence
Elliptic2 large Bitcoin subgraph AML dataset	Suspicious-versus-licit subgraph context; 122,000 labeled subgraphs; context row; fixed partition	pooled moments counts; raw subgraphs local-gated	PR-AUC, P@k, fixed-FPR recall	external reference/context

Modeling effort is intentionally budgeted rather than open-ended. PaySim uses a two-stage competitive budget: probes over a seeded train-only sample, five full-training finalists, validation-only calibration, and one fixed test evaluation for the selected finalist. Elliptic uses a graph-feature budget over source-provided anonymized features, same-snapshot structural features, and their combination. Elliptic2 uses repeated pooled-moment LightGBM context rows with seeds 11, 42, and 73 as an external-reference stress test for the release machinery. The full family/search table is kept in the appendix so the main text can focus on evidence roles rather than hyperparameter inventory.

The feature policy is strictest for PaySim because simulator balance columns can leak post-event state. Relaytic excludes those balance fields, raw origin and destination identifiers as model features, and simulator flags. It allows row-local amount, type, and time features, train-only thresholds, and destination history shifted before each step. The isolated contribution of destination-history features has not been measured as a separate test row, so it is not reported as a main result.

6 Results

Table 4: PaySim modeling path.

Stage	Model/contract	Selection evidence	Final test evidence	Role
P4 reference	SGD logistic baseline	source-safe starting point	0.2159	reference row
P6 baseline	Extra Trees baseline	leakage-safe feature set	0.3313	baseline row
Probe screen	best small-sample probe: XGBoost	26 allowed features; probe validation PR-AUC 0.5944	no test evaluation	candidate screening
Full finalist selection	Extra Trees finalist	full-training validation PR-AUC 0.5687; selected before test	test still hidden	model selection
Final fixed test	Extra Trees with Platt calibration	validation-only calibration and threshold	0.6388	bounded demonstration

PaySim is the most complete local modeling path in the current evidence pack. The earliest reference row was PR-AUC 0.2159. The later leakage-safe baseline improved to 0.3313. The small-sample probe screen then identified a strong XGBoost probe, but fixed-test eligibility was decided later among full-training finalists; Extra Trees had the best full-training validation PR-AUC and was the only competitive finalist evaluated on the fixed test. It reached fixed-test PR-AUC 0.6388 and ROC-AUC 0.9683. The improvement is meaningful inside the synthetic temporal-fraud contract because balance fields were excluded, prior-step destination history was added without raw account encoding, candidates were selected by validation evidence, and calibration and thresholding used validation-only partitions. The admissible interpretation is precise: Relaytic-AML produced a stronger, leakage-audited PaySim temporal-proxy row under a declared budget.

The review-budget metrics sharpen the interpretation. At the selected PaySim review budget, precision is 0.7033 and recall is 0.4716. The fixed queue reviews 1,109 items, containing 780 true positives and 329 false positives; 874 of 1,654 positives remain outside the reviewed set. The top of the queue is much richer than prevalence, but it still misses substantial fraud. The PaySim review-budget row is useful for an evaluation lab because it connects ranking quality to analyst capacity instead of treating PR-AUC as the whole story.

Elliptic is a different kind of evidence. The validation-selected source-plus-structural LightGBM row reports test PR-AUC 0.6688, with review-budget precision 1.0000 and recall 0.0566. The fixed queue reviews 36 items, containing 36 true positives and 0 false positives; 600 of 636 positives remain outside the reviewed set. The Elliptic row supports temporal graph-feature provenance and operating-point reporting. The same row also reveals a limitation: the current graph-structure-only floor is weak, and the final row is heavily influenced by source-provided anonymized features. Relaytic's contribution here is the graph-aware evidence path: feature provenance, temporal splits, operating-point metrics, and interpretation routing are made auditable together.

Elliptic2 is the modern benchmark-context row. The repeated official-partition context row reports PR-AUC 0.9432 +/- 0.0009, and the content-hash robustness partition reports mean PR-AUC 0.9297. Those values sit beside the recorded RevClassifyDS reference of 0.9740, giving the reader a frontier marker while Relaytic carries the external benchmark evidence, cohort notes, and reference-execution requirements as governed context.

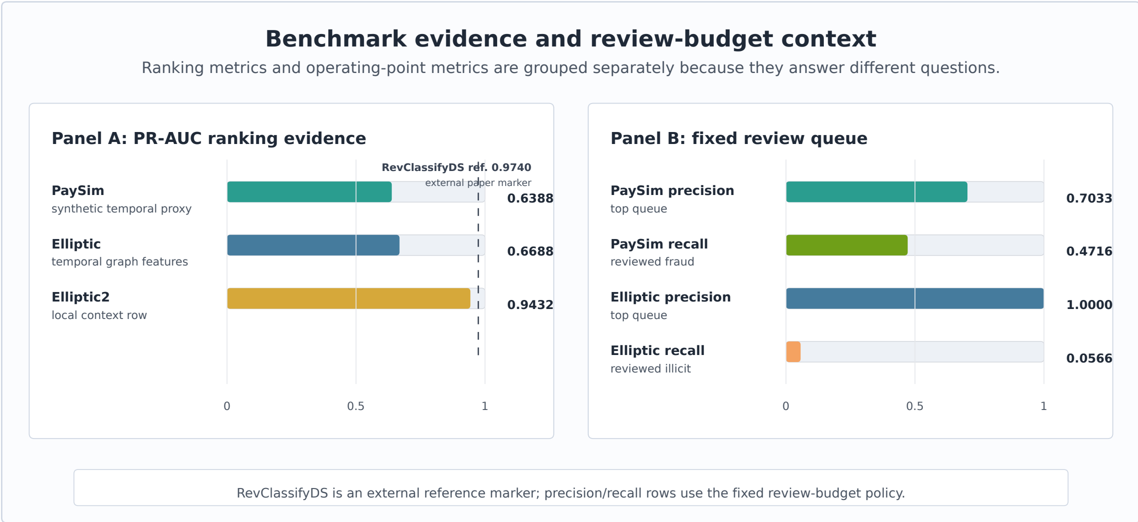

Figure 4: Benchmark and review-budget evidence: PR-AUC is shown beside precision and recall at the bounded review queue instead of being interpreted alone.

Figure 4 separates ranking metrics from operating-point metrics. PR-AUC summarizes ranking quality under rare-event imbalance. Precision and recall at the selected review budget describe what the top of an analyst queue would contain under the paper's fixed policy. Keeping those views together, but visually separated, is important because a useful top queue can still leave many positives unreviewed.

7 System Evaluation

The system claim is evaluated through deterministic reader and agent tasks. These checks ask whether a reviewer can navigate from the README to the paper evidence, trace PaySim metric provenance, compare baseline and competitive budgets, keep Elliptic2 as context rather than a contribution, export rowless state for an external agent, recover an interrupted run, and block over-strong public claims. The main paper reports the synthesis; the detailed failure cases, ablations, invariant map, hosted-score example, and handoff rows are preserved in the appendix and generated evidence artifacts.

Table 5: System evaluation summary.

Check	What can fail	Relaytic mechanism	Observed signal	Boundary
Metric provenance	A reported number cannot be traced to source, split, command, or artifact.	Required evidence-cell fields and metric-cell audit.	13/13 required fields present; metric audit passed.	Proves traceability, not detector optimality.
Budget comparability	Baseline and competitive rows are compared under different contracts.	Dataset, split doctrine, metric, and budget checks.	PaySim PR-AUC improved from 0.3313 to 0.6388 under the same contract.	Supports a bounded PaySim comparison, not SOTA.
Leakage and selection firewall	Post-event fields or test evidence influence the selected model.	Feature policy plus validation-only selection fixtures.	4 forbidden balance fields excluded; no test-set selection; one fixed test finalist.	Benchmark-specific leakage taxonomy.
Claim-strength gating	Proxy or context rows become real-bank, parity, or headline claims.	Public wording lint, publishability matrix, and stronger-claim cases.	Six stronger-claim cases tested; hard and headline claims blocked.	Deterministic release gate, not peer review.
Rowless handoff	An external agent receives raw rows, credentials, or private paths.	Context-export redaction and handoff evaluator.	Rowless handoff preserved next action and allowed tools.	Fixture redaction proof, not a privacy certification.
Interrupted recovery	A user or agent cannot recover current state without artifact literacy.	No-lost-user guide and recovery artifact shortlist.	Recovery guide, partial-run state, and artifact shortlist were emitted.	Deterministic recovery check, not a human study.
Hosted-score wrapper	A third-party score file is mistaken for Relaytic detector novelty.	Schema/hash adapter, evidence cell, redaction report, and claim map.	11 exported fields; 16 blocked fields; no raw rows exported	Hosted detector-output governance only.

The audit summary supports a concrete systems conclusion. Relaytic-AML changes what can be promoted from the evidence pack: a number with missing provenance, a leakage-prone feature path, a test-selected finalist, an unsafe handoff packet, or an over-strong public claim becomes a blocked state rather than publishable wording. These checks evaluate infrastructure behavior; human usability, privacy certification, and production AML validation require separate studies.

The hosted external-score fixture shows the intended integration point for stronger third-party detectors. A rowless detector-score artifact enters Relaytic with schema and content hashes, not raw rows. Relaytic emits one governance evidence cell, redacts unsafe handoff fields, and routes the result as hosted detector-output governance evidence. Future detector outputs can therefore pass through the same local release boundary without being mistaken for a new detector contribution.

8 Limitations and Threats to Validity

PaySim is synthetic. It is useful for controlled temporal fraud experiments, but it is not evidence of bank-scale AML superiority. The simulator has known simplifications, and the current result should be interpreted as a leakage-audited proxy result. The destination-history feature contract is present, but the isolated destination-history ablation is not in the current evidence pack, so no separate result is claimed for that feature family.

Public blockchain data is also not the same as bank AML. Elliptic provides a valuable temporal graph task, but unknown labels, anonymized features, and public-chain behavior limit direct operational interpretation. Elliptic2 is modern and highly relevant, but the current local evidence does not satisfy the stronger reference-parity conditions needed for a performance contribution against RevClassifyDS.

The deterministic system checks are not a substitute for a human usability study. They show that artifacts, redactions, interpretation gates, and recovery surfaces are present and internally consistent. They do not measure analyst time, production incident rates, organizational adoption, or real investigation quality. Future work should test stronger release budgets, repeated runs, private or partner-approved holdouts, same-queue incumbent comparisons, and graph-native families under the same evidence-cell discipline.

The system is intentionally local-first, which creates a tradeoff. Privacy and provenance improve because raw rows stay local, but external reviewers cannot rerun licensed or private data without obtaining it themselves. The paper handles that by publishing commands, hashes where allowed, generated artifacts, and claim boundaries, but a fully independent reproduction of every heavy benchmark still depends on legal access to the source datasets.

9 Reproducibility

The repository is larger than this AML paper. Relaytic is the general local-first inference lab and public package; Relaytic-AML is the focused AML edition used here for the manuscript. A reader should start with the README and this paper. Development-control files record the build history, but they are not required to understand the paper claims. Public citation should use the final release tag or archival snapshot selected at submission time, because the main branch can continue to evolve after the paper is posted.

Repository: https://github.com/ML-Enthusiast-de/Relaytic. Source commit: 7874eef05de1. The arXiv source bundle and preflight reports record artifact hashes for this candidate.

The compact contract below separates what a clean clone can reproduce immediately from what requires local benchmark access. The README contains the full regeneration script; the paper keeps the main path short enough to try without reading the generated audit files first.

Table 6: Reproducibility contract.

Path	Command	Reproduces	Data dependency
Public paper rebuild	invariants; release; polish; novelty; source; final preflight	Markdown, LaTeX source, figures, PDF/log/font preflight	Clean clone with Python >=3.10 and local TeX for PDF checks
System fixtures	paper-system-eval; paper-failure-eval; paper-governance-ablation; paper-invariants	navigation, handoff, recovery, claim-gate, and disabled-component checks	Repo-local deterministic fixtures only
Hosted-score fixture	paper-external-score-proof; paper-external-score-integration	rowless score schema, evidence cell, redaction, and claim map	Repo-local rowless fixture; optional local score files stay local
PaySim benchmark	paysim-competitive –budget-tier competitive –run-optional	selected PaySim PR-AUC and review-budget cells	Local Kaggle PaySim file; sha256 prefix 16910f90577b
Elliptic benchmark	graph-baselines –budget-tier competitive –run-optional	selected graph-feature evidence cells	Local Kaggle Elliptic files; sha256 prefix 93e2e7b2405c plus 2 files
Elliptic2 context	elliptic2-competitive where local access permits	modern benchmark-context row and reference-gate notes	Local Elliptic2/RevTrack artifacts; no parity claim without faithful reference run

Minimal public checks use only repo-local deterministic fixtures and paper artifact-build outputs. Full benchmark regeneration additionally requires local PaySim, Elliptic, and Elliptic2 access where the dataset licenses permit local use but not redistribution.

Minimal public check:

```
python -m pip install -e ".[full]"
python -m relaytic.ui.cli release-safety paper-invariants --format json
python -m relaytic.ui.cli release-safety paper-release --format json
python -m relaytic.ui.cli release-safety paper-narrative-polish --format json
python -m relaytic.ui.cli release-safety paper-novelty-positioning --format json
python -m relaytic.ui.cli release-safety paper-arxiv-source --format json
python -m relaytic.ui.cli release-safety paper-final-preflight --format json
```

Raw benchmark data is not committed. PaySim and Elliptic require local downloads and are referenced through registry artifacts, split reports, hashes, and command ledgers. Elliptic2 is used as benchmark context in this paper because the stronger reference-parity conditions are not satisfied locally. Clean clones can reproduce the paper artifact-build checks and repo-local public fixtures; full benchmark regeneration requires the locally licensed datasets named in the README.

10 AI Assistance Disclosure

Large language model tools assisted with drafting, editing, repository inspection, consistency checks, and implementation work around the paper artifacts. They are not authors. The evidence cells, benchmark outputs, source code, figures, tables, limitations, and final interpretation remain the author's responsibility.

11 Conclusion

Relaytic-AML shows how an agent-assisted AML evaluation lab can be built around local evidence rather than conversational memory. The system keeps data posture, temporal and graph split validity, leakage controls, model budgets, review-budget operating points, rowless handoff, and public claims inside one artifact record. The PaySim, Elliptic, and Elliptic2 rows are useful because they demonstrate that architecture under realistic forms of pressure, including rare events, graph provenance, modern benchmark context, and governed interpretation.

The strongest claim supported today is architectural: Relaytic-AML can make AML experiments easier to inspect, easier to challenge, safer to hand off to another agent, and harder to overstate. Relaytic-AML is best viewed as a governance substrate for stronger future detector studies, not as a replacement for them.

A Detailed Audit and Reproducibility Records

The appendix keeps the concrete audit evidence out of the main reading path while preserving it for reviewers who want to inspect the mechanics. The tables below summarize generated artifacts; the repository stores the corresponding JSON reports with full fields, hashes, and pass criteria.

Table 7: Model families and search budgets.

Track	Families	Features	Search budget	Evidence role
PaySim	tree and boosting candidates; Extra Trees selected	amount, type, time, shifted destination history	14 probes; 5 finalists; seeds 42	validation PR-AUC; one fixed test
Elliptic	tree/boosting baselines; LightGBM selected	source node features plus same-step graph statistics	16 trials; seeds 42	graph-feature evidence row
Elliptic2	LightGBM context row	348 pooled subgraph moments/counts	3 repeated seeds: 11, 42, 73	external reference row

The model-search table records budget shape and evidence role. It is appendix material because it supports auditability without changing the paper's central architectural claim.

Table 8: Detailed failure-case fixtures.

Failure mode	Injected risk	Gate/check	Evidence	Expected behavior	Observed result
Leakage-column injection	PaySim balance fields are offered as candidate model inputs.	Leakage feature policy	PS-PR feature policy	Post-event balance fields stay out of allowed features.	4 offered, 4 excluded, 0 used; labels not used as features
Test-set selection violation	A model-selection path tries to use test evidence before the finalist is fixed.	Validation-only selection policy	PS-PR search contract	Only validation evidence may select, calibrate, or threshold the finalist.	validation-only probes; no test selection; one finalist test
Over-strong claim attempt	Draft wording proposes real-bank superiority or RevClassifyDS parity.	Public claim gate	claim-gate report	Unsupported headline and hard-performance claims remain blocked.	6 blocked claims; hard and headline claims blocked
Rowless handoff redaction	An external-agent packet requests raw rows, private paths, or sensitive fields.	Context export redaction	handoff redaction task	The export contains state and next actions, not raw rows or private paths.	raw rows excluded from export; 8 unsafe fields redacted; 6 blocked fields recorded
Interrupted-run recovery	A user or agent resumes a partial run without knowing which artifact to inspect.	No-lost-user guide	guide recovery task	The guide exposes current state, missing evidence, artifact shortlist, and next actions.	partial run recovered; 8 missing-evidence items recorded; 6 recovery actions exposed

The failure-case fixtures exercise whether the release path refuses leakage features, test-set model selection, over-strong claims, unsafe handoff, and lost-run states. They do not add detector benchmark rows.

Table 9: Governance machinery ablation.

Path	Disabled machinery	Unsafe signal	Artifact integrity	Handoff / recovery	Interpretation
Full governance path	none	0 unsupported claims, leakage inputs, or raw fields	0 missing fields; 3 table groups	6 recovery actions exposed	Claim gate, leakage policy, redaction, provenance, and recovery guide are all active.
No claim gate	public claim gate	6 unsupported claims	3 table groups unchanged	6 recovery actions exposed	The claim gate is what keeps proxy evidence below hard AML, SOTA, RevClassifyDS parity, production, and business-value claims.
No leakage policy	PaySim feature leakage policy	4 leakage inputs	0 missing fields; 1 unsafe table path	6 recovery actions exposed	The leakage policy prevents post-event simulator fields from becoming apparently strong evidence.
No rowless handoff redaction	external-agent redaction	6 raw fields	3 table groups unchanged	6 raw fields; 6 recovery actions exposed	Rowless handoff is the privacy boundary that lets outside agents help without receiving raw data or local paths.
No evidence-cell required fields	metric-cell required-field gate	13 missing provenance fields	13 missing fields; 1 unsafe table path	6 recovery actions exposed	Required fields are what connect a reader-facing number back to dataset, split, command, artifact, leakage, budget, and claim state.
No interrupted-run recovery guide	no-lost-user guide	0 recovery actions	3 table groups unchanged	0 recovery actions exposed	The recovery guide is what keeps state navigation from depending on repo literacy.

The governance ablation compares the full path with disabled-component fixtures. The detector scores do not change; what changes is which wording and handoff surfaces are allowed to leave the evidence pack.

Table 10: Governance invariants and evidence map.

Invariant	Mechanism	Evidence and stress signal	Boundary
Metric-cell provenance	metric-cell audit plus required-field gate	audit: pass; stress ablation: 13 provenance fields missing; release blocked	The invariant proves artifact completeness, not that the detector is optimal.
Claim-strength monotonicity	claim lint, allowed-claims report, publishability matrix, and overclaim failure case	claim lint: pass; stress overclaim stress: 6 unsupported claims blocked	The gate is a deterministic release check; it is not an external peer review.
Leakage and selection firewall	feature policy report, split contract, failure fixtures, and leakage ablation	leakage policy: 4 forbidden fields offered; 0 used; stress leakage stress: 4 leakage fields offered and excluded; 0 used	The current firewall is benchmark-specific; future datasets need their own leakage taxonomy.
Rowless external-agent handoff	handoff evaluator plus redaction failure case	handoff audit: raw rows excluded; 8 unsafe fields redacted; 6 fields blocked; stress redaction stress: 6 unsafe fields blocked; raw rows excluded	The check proves deterministic redaction on fixtures, not a broad privacy certification.
Interrupted-run recoverability	no-lost-user guide and partial-run recovery fixtures	recovery audit: partial run recovered; 8 missing items; 6 actions exposed; stress recovery stress: partial run recovered; 6 actions exposed	The check is deterministic; it does not measure human time-to-recovery.
Benchmark role separation	publishability matrix and allowed-claims report	role matrix: 5 supporting rows allowed; hard/headline claims blocked; stress claim stress: 6 public claims blocked	Rows with external or proxy roles cannot be treated as unified leaderboard evidence.
Local-first release safety	release go/no-go, claim lint, and public-claim whitelist	wording lint: pass; stress claim boundary: headline and hard performance claims blocked	Licensed benchmark files are not redistributed; reproduction depends on local access.

The invariant map records release-time rules rather than prose preferences. Each invariant pairs a mechanism with an observed stress signal and an explicit boundary.

Table 11: Hosted external-score case study.

Component	Observed evidence	Evidence	Admissible interpretation
Adapter input	external-score adapter over a rowless fixture; schema hash 4b2b70a58b0c; content hash dac68c3801f5	schema/hash report	The score artifact is described by schema and hash posture, not by raw rows.
Evidence emitted	1 evidence cell; metadata-completeness metric; value 1.0000	evidence-cell report	Relaytic records the governance metric as auditable evidence, not as detector novelty.
Rowless handoff	11 exported fields; 16 blocked fields; no raw rows exported	handoff-redaction report	A downstream agent can inspect state without receiving rows, identifiers, paths, or secrets.
Claim state	hosted detector-output governance only; 5 stronger claims blocked	claim-gate report	The public use is hosted detector-output governance only.


```
{
  "cell_id": "p19a.external_score.hosted_metadata_completeness",
  "dataset_id": "p19a_hosted_score_fixture",
  "split": "fixture_holdout",
  "command": "relaytic release-safety paper-external-score-proof",
  "artifact_ref": "fixture:p19a_external_score_rowless_v1",
  "metric": "hosted_score_metadata_completeness",
  "value": 1.0,
  "leakage_posture": "rowless_no_training_or_label_data_exported",
  "claim_state": "hosted_detector_output_governance_only"
}
```


The hosted-score record is metadata governance only. It proves that a rowless score artifact can be wrapped by schema, hash, redaction, and claim-state records.

Table 12: Evidence routing examples.

Stronger future use	Current admissible use	Evidence needed
Real-bank deployment study	bounded PaySim temporal-proxy demonstration	Partner or bank-approved holdout, incumbent comparison, analyst-review protocol, and legal release gate.
Elliptic2 reference-method comparison	external RevClassifyDS reference marker plus local context row	Faithful reference execution, cohort reconciliation, resource budget, and repeated parity report.
Graph-native detector release	Elliptic temporal graph-feature evidence path	Graph-native release budget, neural baselines, repeated seeds, and graph-specific ablations.

The blocked-claim rows show how stronger future uses are handled. The gate records current admissible use and the evidence needed before a stronger interpretation could be made.

Table 13: Rowless handoff and interrupted-run recovery examples.

Scenario	Input state	Exported fields	Redacted fields	Observed signal
External-agent handoff	partial run with available guide state	state summary, action options, starter questions, tool contract, artifact shortlist	raw transaction rows, credentials, private paths, raw source files	raw rows excluded from export; 8 unsafe fields redacted; 6 blocked fields recorded
Safe next action	external model asked what to do next	six next actions, six starter questions, command options	unredacted local paths and data rows	6 next actions and 6 starter questions exposed
Interrupted-run recovery	operator returns to partial run without artifact literacy	current state, missing evidence count, canonical artifact shortlist, context-export command	raw benchmark data and private machine paths	partial run recovered; 8 missing-evidence items and 6 actions exposed

The handoff and recovery rows give the practical external-agent story. A second model can receive state, commands, artifacts, and starter questions, while raw rows remain redacted and private paths stay withheld.

Appendix reproduction shortcut:

The full Windows and macOS/Linux regeneration script is kept in the README so the appendix remains readable. The essential local paper path is:

Windows PowerShell:

```
py -3.11 -m pip install -e ".[full]"
py -3.11 -m relaytic.ui.cli release-safety paper-invariants --format json
py -3.11 -m relaytic.ui.cli release-safety paper-release --format json
py -3.11 -m relaytic.ui.cli release-safety paper-narrative-polish --format json
py -3.11 -m relaytic.ui.cli release-safety paper-novelty-positioning --format json
py -3.11 -m relaytic.ui.cli release-safety paper-arxiv-source --format json
py -3.11 -m pytest tests/test_paper_track_p20.py tests/test_paper_track_p21.py
↪  tests/test_paper_track_p23.py -q
```

After compiling `docs/paper/arxiv_src/main.tex` and copying the PDF to the review draft, run `paper-final-preflight`. The README includes the exact compile/copy commands and the longer audit test matrix.

macOS/Linux:

```
python3 -m pip install -e ".[full]"
python3 -m relaytic.ui.cli release-safety paper-invariants --format json
python3 -m relaytic.ui.cli release-safety paper-release --format json
python3 -m relaytic.ui.cli release-safety paper-narrative-polish --format json
python3 -m relaytic.ui.cli release-safety paper-novelty-positioning --format json
python3 -m relaytic.ui.cli release-safety paper-arxiv-source --format json
python3 -m pytest tests/test_paper_track_p20.py tests/test_paper_track_p21.py
↪  tests/test_paper_track_p23.py -q
```

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